

ENFLOXIN ORAL SOLUTION 100MG/ML

Ingredient(s):

Each ml contains :

Enrofloxacin 100mg

Pharmacology (Summary of Pharmacodynamics and Pharmacokinetics):

Enrofloxacin is a fluoroquinolone antibiotic and is a bactericidal agent. The bactericidal activity of enrofloxacin is concentration dependent, with susceptible bacteria cell death occurring within 20–30 minutes of exposure. Enrofloxacin has demonstrated a significant post-antibiotic effect for both G(-) and G(+) bacteria and is active in both stationary and growth phases of bacterial replication.

Its mechanism of action is believed to act by inhibiting bacterial DNA-gyrase (a type-II topoisomerase), thereby preventing DNA supercoiling and DNA synthesis.

Enrofloxacin is well absorbed after oral administration in most species. In dogs, bioavailability is approximately 80%. 50% of Cmax is reportedly attained within 15 minutes of dosing and peak levels (Cmax) occur within one hour of dosing. The presence of food in the stomach may delay the rate, but not the extent of absorption. In sheep, enrofloxacin administered orally is about 65–75% bioavailable.

Enrofloxacin is distributed throughout the body. Volume of distribution in dogs is approximately 3–4L/kg. Only about 27% is bound to canine plasma protein. Highest concentrations are found in the bile, kidney, liver, lungs, and reproductive system (including prostatic fluid and tissue). Enrofloxacin reportedly concentrates in macrophages. Therapeutic levels are also attained in bone, synovial fluid, skin, muscle, aqueous humor and pleural fluid. Low concentrations are found in the CSF, and levels may only reach 6–10% of those found in the serum. In cattle, the volume of distribution is about 1.5L/kg and in sheep, 0.4L/kg.

Enrofloxacin is eliminated via both renal and non-renal mechanisms. Approximately 15–50% of the drug is eliminated unchanged into the urine, by both tubular secretion and glomerular filtration. Enrofloxacin are metabolized to various metabolites that are less active than the parent compounds. Approximately 10–40% of circulating enrofloxacin is metabolized to ciprofloxacin in most species, including humans, dogs, cats, adult horses, cattle, turtles, and snakes. Foals, pigs, and some lizards apparently do not convert much enrofloxacin, if any, to ciprofloxacin. These metabolites are eliminated both in the urine and feces. Because of the dual (renal and hepatic) means of elimination, patients with severely impaired renal function may have slightly prolonged half-lives and higher serum levels. The approximate elimination half-lives in various species are: dogs 4–5 hours; cats 6 hours; sheep 1.5–4.5 hours; horses 6 hours; turtles 18 hours; and alligators 55 hours.

Indication(s):

For use in the treatment of diseases of the respiratory and alimentary tracts of bacterial or mycoplasmal origin (e.g. pasteurellosis, mycoplasmosis, coli-bacillosis, coli-septicaemia and salmonellosis) where clinical experience, supported where possible by sensitivity testing of the causal organism, indicates enrofloxacin as the drug of choice in chicken and turkeys.

Target Species

Chicken, turkey

Dosage and Administration:

The product is administered via the drinking water. Medicated water should be made up immediately prior to provision. Medication of the water supply should be continuous during the treatment period and no other source of water should be available. The product should be given over a minimum period of 6 hours per day to allow all birds to drink.

Chickens and Turkeys

The dose rate is 10 mg/kg bodyweight daily for 3 to 10 days.

Each ml contains 100 mg enrofloxacin.

Calculate the daily quantity (ml) of the product required for treatment period as follows :

Total number of birds x Average body weight in kg x 0.1 = Total volume (ml) per day.

The product may be put directly into the header tank or introduced via a water proportioner pump.

For the treatment of salmonellosis or pasteurellosis a treatment period of 5 to 10 days must be used. For other conditions the duration of treatment may be 3 to 5 days.

Contraindications:

- 1) Enrofloxacin is contraindicated in small and medium breed dogs from 2 months to 8 months of age. Bubble-like changes in articular cartilage have been noted when the drug was given at 2–5 times recommended doses for 30 days, although clinical symptoms have only been seen at the 5X dose. Large and giant breed dogs may be in the rapid-growth phase for periods longer than 8 months of age, so longer than 8 months may be necessary to avoid cartilage damage.
- 2) Quinolones are also contraindicated in patients hypersensitive to them.
- 3) The product should not be used for prophylaxis.

Precaution(s) / Warning(s):

- 1) Because ciprofloxacin has occasionally been reported to cause crystalluria in humans, animals should not be allowed to become dehydrated during therapy with either ciprofloxacin or enrofloxacin.
- 2) Enrofloxacin may cause CNS stimulation and should be used with caution in patients with seizure disorders.
- 3) Patients with severe renal or hepatic impairment may require dosage adjustments to prevent drug accumulation.
- 4) Enrofloxacin should not be used by humans as it may cause hallucinations, vivid dreams, and headache.

Interaction with Other Medicaments

- 1) Antacids containing cations (Mg^{++} , Al^{+++} , Ca^{++}) may bind to enrofloxacin and prevent its absorption.
- 2) Sucralfate may inhibit absorption of enrofloxacin, separate doses of these drugs by at least 2 hours.
- 3) Enrofloxacin administered with theophylline may increase theophylline blood levels.
- 4) Probenecid blocks tubular secretion of enrofloxacin and may increase its blood level and half-life.
- 5) Synergism may occur, but is not predictable, against some bacteria (particularly *Pseudomonas aeruginosa* or other *Enterobacteriaceae*) with enrofloxacin and aminoglycosides, 3rd generation cephalosporins agents, and extended-spectrum penicillins.
- 6) Although enrofloxacin has minimal activity against anaerobes, in vitro synergy has been reported when used with clindamycin against strains of *Peptostreptococcus*, *Lactobacillus* and *Bacteroids fragilis*.
- 7) Nitrofurantoin may antagonize the antimicrobial activity of the fluoroquinolones and their concomitant use is not recommended.
- 8) Fluoroquinolones may exacerbate the nephrotoxicity of cyclosporine (used systemically).

Pregnancy and Lactation

- Pregnancy and lactation : Not applicable, the product is indicated only for use in poultry.
- Lay : Do not administer to layer replacement birds within 14 days of coming into lay.

Side Effect(s) / Adverse Reaction(s):

- 1) With the exception of potential cartilage abnormalities in young animals, the adverse effect profile of enrofloxacin is usually limited to GI distress (vomiting, anorexia).
- 2) In dogs, rare incidences of elevated hepatic enzymes, ataxia, seizures, depression, lethargy and nervousness have also been reported. Hypersensitivity reactions or crystalluria could potentially occur.
- 3) In cats, rare incidences of ocular toxicity have been reported characterized by mydriasis, retinal degeneration and blindness. These effects were generally seen at higher dosage ranges ($>15mg/kg$) and have necessitated a reduction in dosage recommendations in cats to a maximum of $5mg/kg/day$. Other rare adverse effects seen in cats may include vomiting, anorexia, elevated hepatic enzymes, diarrhea, ataxia, seizures, depression/lethargy, vocalization and aggression.

Environmental Property

No data available for this product.

Symptoms and Treatment for Overdosage, and Antidote(s):

An acute overdose of enrofloxacin would result in either anorexia or vomiting. Dogs receiving 10 times the labeled rate of enrofloxacin for at least 14 days developed only vomiting and anorexia. Death did occur in some dogs when feed 25 times the labeled rate for 11 days.

Shelf-Life:

3 years.

Storage Condition(s):

Store below $30^{\circ}C$.

Product Description and Packing(s):

A clear yellow solution.
Plastic bottle of 1L

For food producing animals product:**Withdrawal Periods**

Do not administer to layer replacement birds within 14 days of coming into lay.

- Chicken: Meat: 8 days
Eggs: Contraindicated (not for use in birds producing eggs for human consumption)
- Turkey: Meat: 8 days

Maximum Residual Limit (MRL):

Bovine, ovine, caprine: Muscle $100\mu g/kg$, fat $100\mu g/kg$, liver $300\mu g/kg$, kidney $200\mu g/kg$, milk $100\mu g/kg$.

Porcine, rabbits: Muscle $100\mu g/kg$, fat $100\mu g/kg$, liver $200\mu g/kg$, kidney $300\mu g/kg$.

Poultry: Muscle $100\mu g/kg$, skin + fat $100\mu g/kg$, liver $200\mu g/kg$, kidney $300\mu g/kg$.

All food producing species except bovine, ovine, caprine, porcine, rabbits and poultry: Muscle $100\mu g/kg$, fat $100\mu g/kg$, liver $200\mu g/kg$, kidney $200\mu g/kg$.



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